

# Linear Regression with One Regressor

Chapter 4

ECON3500: Econometrics and Applications

2026-01-01

# Learning Roadmap

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## The Central Question

How do we move from observing **relationships** in data to making **predictions** and testing **causal claims**?

## By the end of this chapter, you will be able to:

- Set up and estimate a simple linear regression model
- Interpret slopes and intercepts with precision
- Calculate and understand fitted values and residuals
- Assess model quality using  $R^2$ , TSS, ESS, SSR, and SER
- State the key assumptions behind OLS estimation
- Implement everything in Stata

# Our Journey

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1. The linear regression model
2. Estimating coefficients: ordinary least squares
3. Measuring how well the model fits
4. Assumptions for causal interpretation
5. Sampling distributions and uncertainty
6. Common misconceptions and clarifications
7. Implementing OLS in Stata
8. Bringing it all together

# The Linear Regression Model

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# Motivating Question

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## Returns to education

What are the economic returns to education?

### Why this matters:

- We observe that more educated people tend to earn more
- Is this just correlation, or something more?
- How much more do we expect someone with 16 years of education to earn vs. someone with 12 years?
- Can we **quantify** the relationship systematically?

**Our goal:** Move from vague observations to a **prediction rule** we can compute from data.

# Data Source

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**Example data:** Education and Annual Wages

**Source:** American Community Survey (ACS) 2024

**Sample:** 300 observations

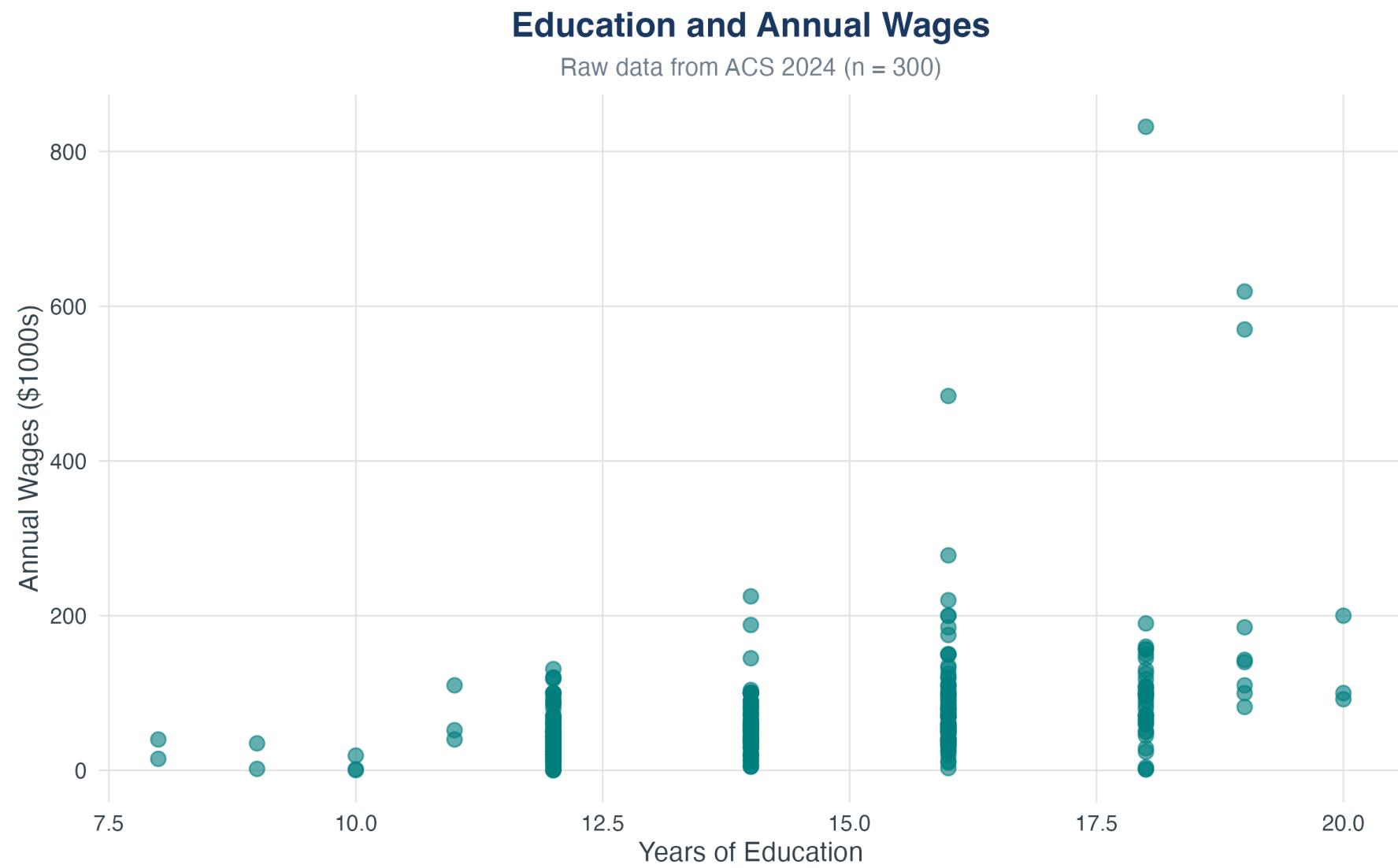
**Variables:**

**Education:** Years of schooling (8-20 years)

**Wages:** Annual earnings in thousands of 2024 dollars

*Note: Based on realistic patterns from ACS microdata*

# First Look at the Data



What do you see?

# From Picture to Model

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Three tools for describing relationships:

1. **Scatter plots** give us a first look
2. **Correlation** tells us direction and strength
3. **Linear regression** gives us a mathematical summary we can use for prediction

## ! Key Insight

Regression transforms a cloud of points into an equation. That equation becomes a **model** we can interrogate, test, and use.



# What Regression Analysis Does

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## Regression helps us:

- **Predict:** Given someone's education, what wages would we expect?
- **Explain:** How does a one-year change in education map to wage changes?
- **Control:** What's the education-wages relationship *holding other factors fixed*?

## Vocabulary:

- **Dependent variable** ( $Y$ ): the outcome we want to explain (wages)
- **Independent variable** ( $X$ ): the driver we use to explain it (education)

# The Population Model

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In the population, the true relationship is:

$$Y = \beta_0 + \beta_1 X + u$$

Components:

- $\beta_0$ : **intercept** — baseline level of  $Y$  when  $X = 0$
  - $\beta_1$ : **slope** — change in  $Y$  from a one-unit increase in  $X$
  - $u$ : **error term** — all other influences on  $Y$  not captured by  $X$
- This is a *model* — a simplified representation of reality. Our job: estimate  $\beta_0$  and  $\beta_1$  from data.

# Understanding the Error Term

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$$\text{Wages} = \beta_0 + \beta_1 \cdot \text{Education} + u$$

## What's in $u$ ?

- Ability and talent
- Work experience
- Gender
- Family background
- Pure luck
- Measurement error

### Critical Point

We can't observe  $u$ , but its properties determine whether we can interpret  $\beta_1$  **causally**. More on this later.

# Ceteris Paribus: Holding Other Things Fixed

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*Ceteris paribus* = “other things equal”

$$Y = \beta_0 + \beta_1 X + u$$

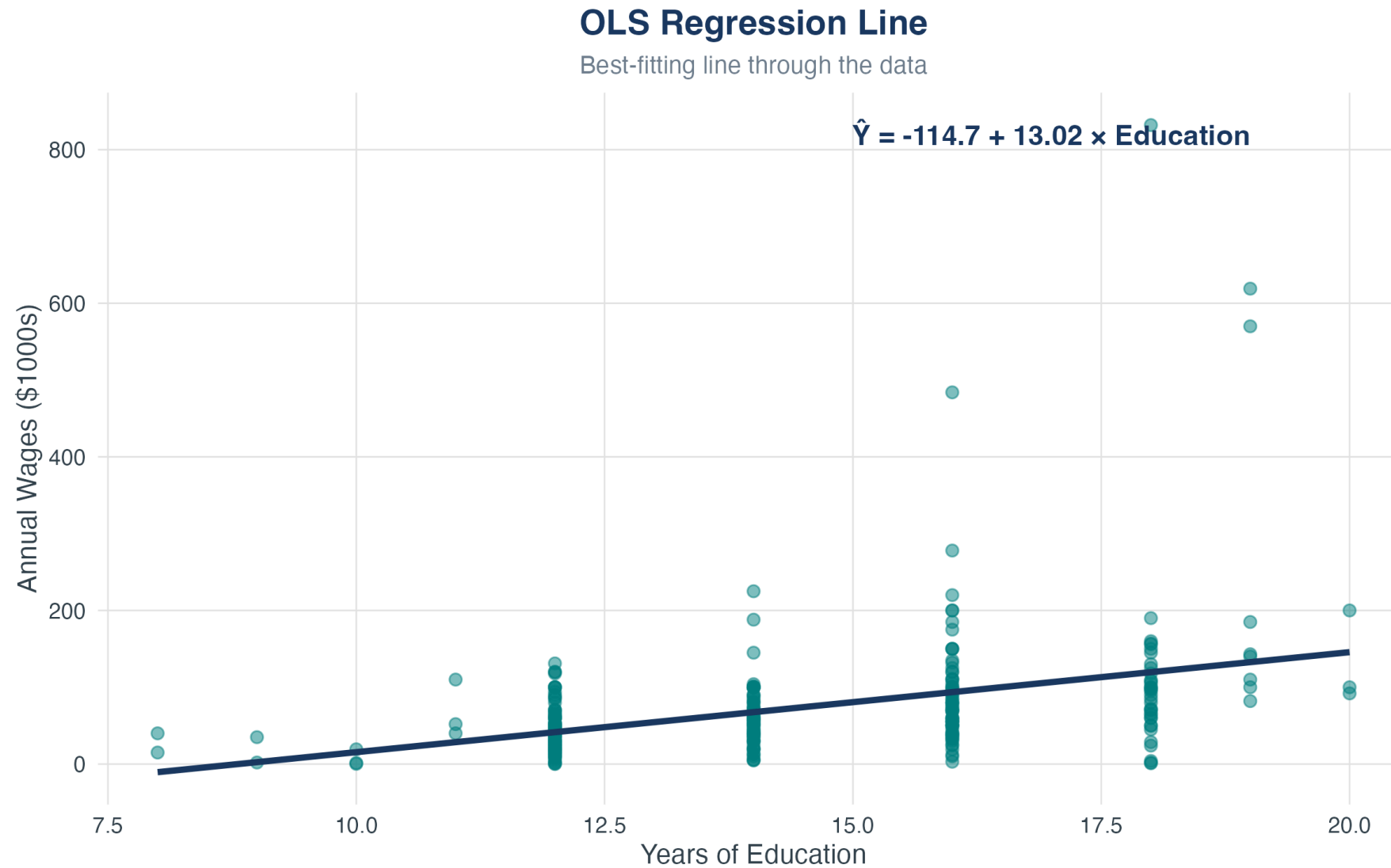
**Taking changes:**

$$\Delta Y = \beta_1 \Delta X + \Delta u$$

**Holding  $u$  fixed means  $\Delta u = 0$ :**

$$\Delta Y = \beta_1 \Delta X$$

# The Regression Line in Action



The line is:  $\hat{Y} = \beta_0 + \beta_1 X$

# Estimating Coefficients: Ordinary Least Squares

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# From Population to Sample

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## The challenge:

- In the population:  $Y_i = \beta_0 + \beta_1 X_i + u_i$
- We observe  $(X_i, Y_i)$  for  $i = 1, \dots, n$
- We do **not** observe  $u_i$
- We must *estimate*  $\beta_0$  and  $\beta_1$  from the sample

## Notation:

- $\hat{\beta}_0, \hat{\beta}_1$  = our estimates (the “hats” indicate estimates)
- Goal: Make these estimates as close as possible to the true  $\beta_0, \beta_1$

# The OLS Criterion

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**Idea:** Choose  $\hat{\beta}_0$  and  $\hat{\beta}_1$  to make prediction errors as small as possible.

**For each observation, the prediction error is:**

$$Y_i - (\beta_0 + \beta_1 X_i)$$

**OLS minimizes the sum of squared errors:**

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)]^2$$

**Why square?**

- Positive and negative errors don't cancel out
- Penalizes large mistakes more than small ones
- Leads to a closed-form solution (calculus magic!)



# Deriving the OLS Estimators [Optional]

## For Math Lovers

This derivation uses calculus to show where the OLS formulas come from. Feel free to skip if you prefer!

**Setup:** Minimize the sum of squared residuals

$$S(\beta_0, \beta_1) = \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)]^2$$

**Step 1: Take first-order conditions (FOCs)**

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_i)] = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n X_i [Y_i - (\beta_0 + \beta_1 X_i)] = 0$$

# Deriving the OLS Estimators (continued)

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Step 3: Substitute  $\hat{\beta}_0$  into the second FOC

$$\begin{aligned}\sum X_i Y_i &= \hat{\beta}_0 \sum X_i + \hat{\beta}_1 \sum X_i^2 \\ \sum X_i Y_i &= (\bar{Y} - \hat{\beta}_1 \bar{X}) \sum X_i + \hat{\beta}_1 \sum X_i^2 \\ \sum X_i Y_i &= \bar{Y} n \bar{X} - \hat{\beta}_1 n \bar{X}^2 + \hat{\beta}_1 \sum X_i^2\end{aligned}$$

Step 4: Solve for  $\hat{\beta}_1$

$$\begin{aligned}\hat{\beta}_1 (\sum X_i^2 - n \bar{X}^2) &= \sum X_i Y_i - n \bar{X} \bar{Y} \\ \hat{\beta}_1 &= \frac{\sum X_i Y_i - n \bar{X} \bar{Y}}{\sum X_i^2 - n \bar{X}^2} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}\end{aligned}$$

**Result:** This is the formula for  $\hat{\beta}_1$  in “The OLS Formulas” slide!

# The OLS Formulas

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**Solution from calculus (see previous slides for derivation):**

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

**Intuition:**

- $\hat{\beta}_1$  depends on how  $X$  and  $Y$  covary
- Need variation in  $X$  to estimate  $\beta_1$  (if all  $X_i$  are the same, we can't learn about the slope!)

# Fitted Values and Residuals

Once we have  $\hat{\beta}_0$  and  $\hat{\beta}_1$ :

## $i$ Fitted Value

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

This is the model's prediction for observation  $i$ .

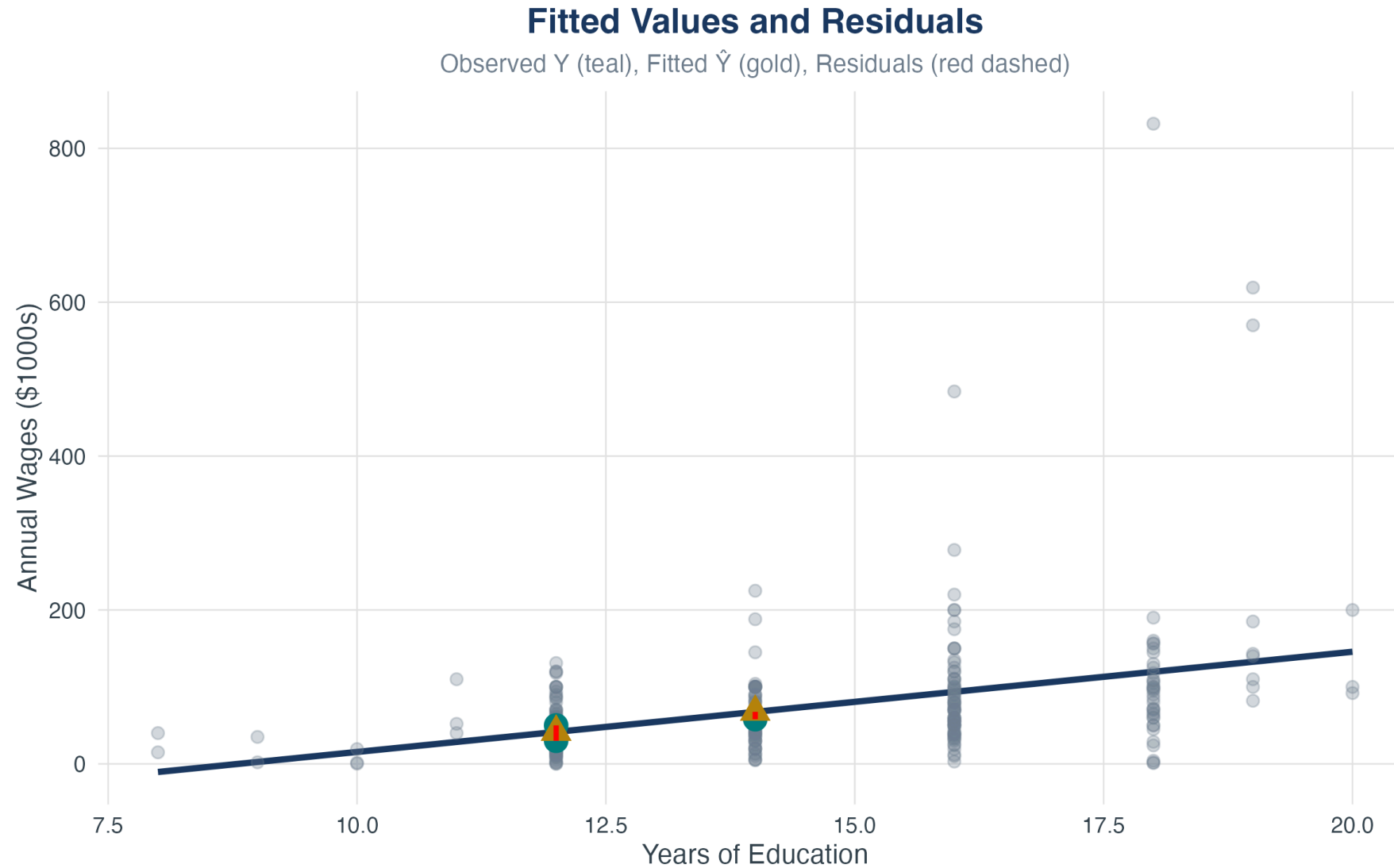
## $i$ Residual

$$\hat{u}_i = Y_i - \hat{Y}_i$$

This is the prediction error for observation  $i$ .

► **Key property:**  $\sum_{i=1}^n \hat{u}_i = 0$  (residuals always sum to zero)

# Visualizing Residuals



- Gold lines: residuals (vertical distance from point to line)

# Stata: Running OLS Regression

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## Basic command:

```
regress wages education
```

## Output includes:

- Coefficient estimates:  $\hat{\beta}_0$  (\_cons) and  $\hat{\beta}_1$  (education)
- Standard errors (for inference - we'll cover this later)
- $R^2$  (measure of fit)
- Number of observations

## Generate fitted values and residuals:

```
predict wages_fitted, xb  
predict residual, residuals
```

# Reading Stata Regression Output

## Understanding the table:

```
Source | SS df MS Number of obs = 300
-----+----- F(1, 298) = 54.60
Model | 298054 1 298054 Prob > F = 0.0000
Residual | 1626752 298 5459.23 R-squared = 0.155
-----+----- Adj R-squared = 0.152
Total | 1924806 299 6436.81 Root MSE = 73.883

-----+-----
wages | Coef. Std. Err. t P>|t| [95% Conf. Interval]
-----+-----
education | 13.022 1.762 7.39 0.000 9.552 16.492
_cons | -114.740 25.633 -4.48 0.000 -165.187 -64.292
-----+-----
```

- **Coef.** column:  $\hat{\beta}_1 = 13.02$ ,  $\hat{\beta}_0 = -114.7$
- **R-squared**: 0.155 (model explains 15.5% of variation)
- **Root MSE**: Standard error of regression (SER) = 73.88
- **Std. Err.**: We'll use this for hypothesis tests (next chapter!)

# Interpreting the Output

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Example results:

$$\widehat{\text{Wages}} = -114.7 + 13.02 \cdot \text{Education}, \quad N = 300$$

Interpretations:

- **Slope (13.02):** Each additional year of education is associated with approximately \$13,020 more in annual wages
- **Intercept (-114.7):** Predicted wages when education = 0 (not meaningful here!)
- **Predictions:**
  - Person with 16 years of education:  $\hat{Y} = -114.7 + 13.02(16) = 93.6\text{k}$
  - Person with 12 years of education:  $\hat{Y} = -114.7 + 13.02(12) = 41.5\text{k}$
- **Residual example:** If someone with 12 years of education earns \$35k, their residual is  $40 - 41.5 = -6.5\text{k}$



# Measuring How Well the Model Fits

# Decomposing Variation

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For any observation:

$$Y_i = \hat{Y}_i + \hat{u}_i$$

We can split total variation into two parts:

$$\underbrace{Y_i - \bar{Y}}_{\text{Total deviation}} = \underbrace{\hat{Y}_i - \bar{Y}}_{\text{Explained deviation}} + \underbrace{\hat{u}_i}_{\text{Unexplained deviation}}$$

Squaring and summing across all observations:

$$\underbrace{\sum_{i=1}^n (Y_i - \bar{Y})^2}_{\text{TSS}} = \underbrace{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}_{\text{ESS}} + \underbrace{\sum_{i=1}^n \hat{u}_i^2}_{\text{SSR}}$$

# The Sum of Squares Trinity

## i Total Sum of Squares (TSS)

$$TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

Measures total variation in  $Y$  around its mean.

## i Explained Sum of Squares (ESS)

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

Variation in  $Y$  explained by the model (variation in fitted values).

## i Sum of Squared Residuals (SSR)

$$SSR = \sum_{i=1}^n u_i^2$$

Variation in  $Y$  **not** explained by the model (variation in residuals).

# Why Does TSS = ESS + SSR? (The Algebra)

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Start with the decomposition:

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i)$$

Square both sides:

$$(Y_i - \bar{Y})^2 = [(\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i)]^2$$

Expand:

$$(Y_i - \bar{Y})^2 = (\hat{Y}_i - \bar{Y})^2 + (Y_i - \hat{Y}_i)^2 + 2(\hat{Y}_i - \bar{Y})(Y_i - \hat{Y}_i)$$

Sum over all  $i$ :

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + 2 \sum_{i=1}^n (\hat{Y}_i - \bar{Y})(Y_i - \hat{Y}_i)$$

# Visualizing the Decomposition

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# R-Squared: The Fraction Explained

  $R^2$  (R-squared)

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}$$

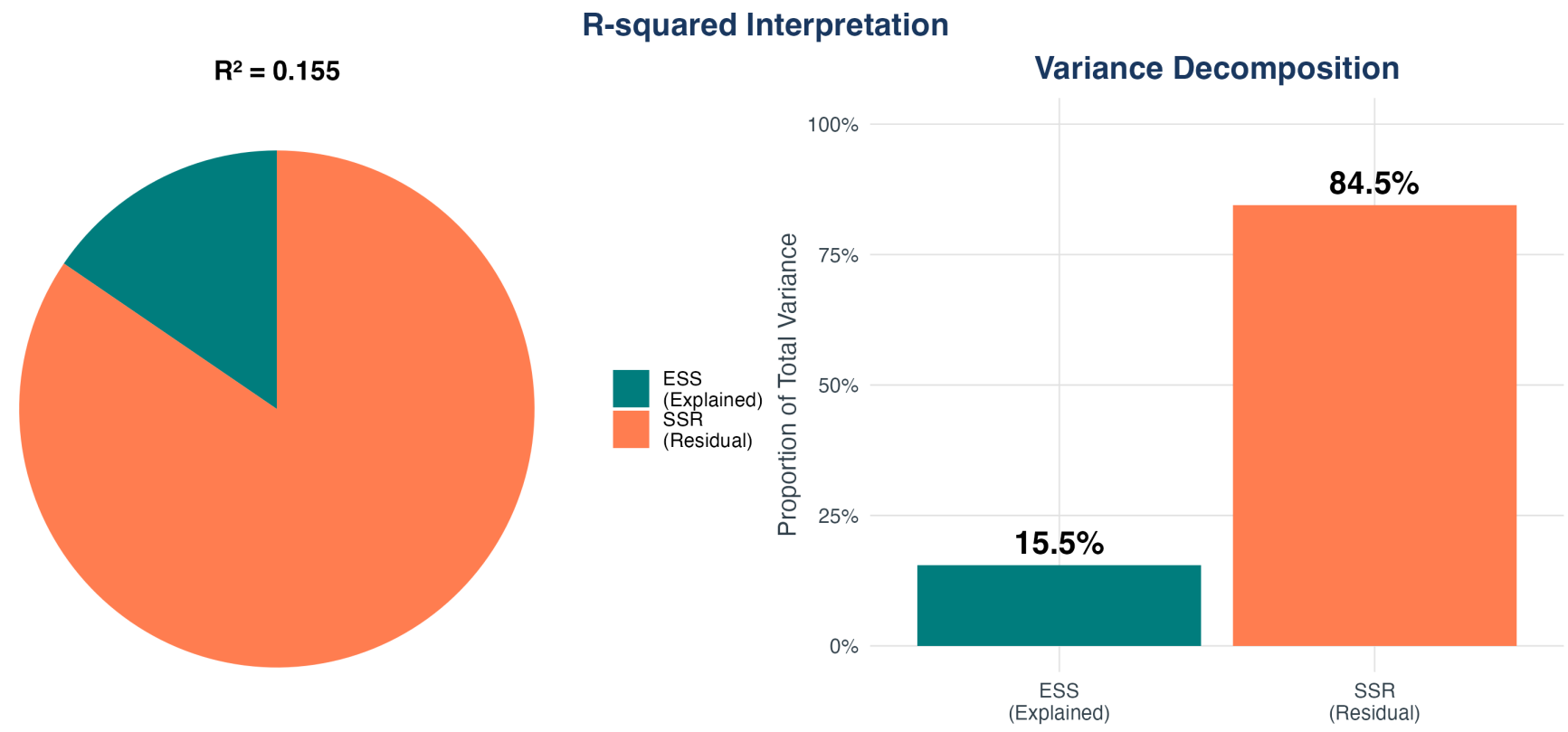
## Properties:

- $0 \leq R^2 \leq 1$
- $R^2 = 0$ : The model explains none of the variation (horizontal line at  $\bar{Y}$ )
- $R^2 = 1$ : The model explains all variation (perfect fit, all points on the line)
- Typical values:  $R^2 = 0.10$  to  $0.30$  in social sciences (many factors matter!)

## Critical Warning

A high  $R^2$  does **not** mean:

# R-Squared Visualization



# Standard Error of the Regression (SER)

How big are the typical residuals?

 Standard Error of Regression

$$SER = \sqrt{\frac{SSR}{n-2}} = \sqrt{\frac{\sum_{i=1}^n u_i^2}{n-2}}$$

Also called the “root mean squared error” (RMSE).

## Interpretation:

- Measures typical size of prediction errors
- In original units of  $Y$  (unlike  $R^2$  which is unitless)
- Lower SER = better fit
- Divide by  $n - 2$  (not  $n$ ) because we estimated 2 parameters

**Example:** If  $SER = 7.9$  in our education-wages regression, typical prediction error is about \$7,900.



# Assumptions for Causal Interpretation

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# A Cautionary Tale: High $R^2 \neq$ Causality

## ⚠️ Provocative Example

Suppose we regress **drowning deaths** on **ice cream sales**:

$$\widehat{\text{Drownings}} = 2.1 + 0.003 \cdot \text{Ice Cream Sales}$$

$R^2 = 0.89$  (very high fit!)

**Does ice cream *cause* drowning?**

► **No!** Both are caused by a third factor: **summer weather**

- Hot days  $\Rightarrow$  more ice cream sales
- Hot days  $\Rightarrow$  more swimming  $\Rightarrow$  more drownings

**The lesson:** A model can fit the data well (high  $R^2$ ) but still fail to identify a causal effect. This is why we need **assumptions**.

# Why Assumptions Matter

## ! The Fundamental Question

We've estimated  $\hat{\beta}_1$ , but does it estimate the **causal effect** of  $X$  on  $Y$ , or just an association?

To interpret  $\beta_1$  causally, we need assumptions.

Three key assumptions for unbiased OLS:

1. **Zero conditional mean:**  $E[u|X] = 0$
2. **i.i.d. sampling:**  $(X_i, Y_i)$  are independent and identically distributed
3. **Finite fourth moments:** Large outliers are unlikely

# Assumption 1: Zero Conditional Mean

## Zero Conditional Mean

$$E[u|X] = 0 \quad \text{for all values of } X$$

The expected value of the error term is zero, **regardless of**  $X$ .

### What this means:

- On average,  $u$  is unrelated to  $X$
- Knowing  $X$  doesn't help you predict  $u$
- All the systematic relationship between  $X$  and  $Y$  is captured by  $\beta_0 + \beta_1 X$

## This is the KEY assumption

If  $E[u|X] \neq 0$ , then  $\hat{\beta}_1$  is **biased** — it systematically estimates the wrong thing. This is called **omitted variable bias**.

# Visualizing Zero Conditional Mean

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# When Does Zero Conditional Mean Fail?

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## Example: Education and wages

$$\text{Wage} = \beta_0 + \beta_1 \cdot \text{Education} + u$$

### What's in $u$ ?

- Ability, motivation, family background, connections

### Problem:

- High-ability people tend to get more education
- So  $E[u|\text{Education} = \text{high}] > E[u|\text{Education} = \text{low}]$
- Violates zero conditional mean!
- Result:  $\hat{\beta}_1$  overestimates the causal effect of education (it picks up both education *and* ability)

# Building Intuition: Mean Independence

A weaker condition than zero conditional mean:

## Mean Independence

$$E[u|X] = E[u] \quad \text{for all } X$$

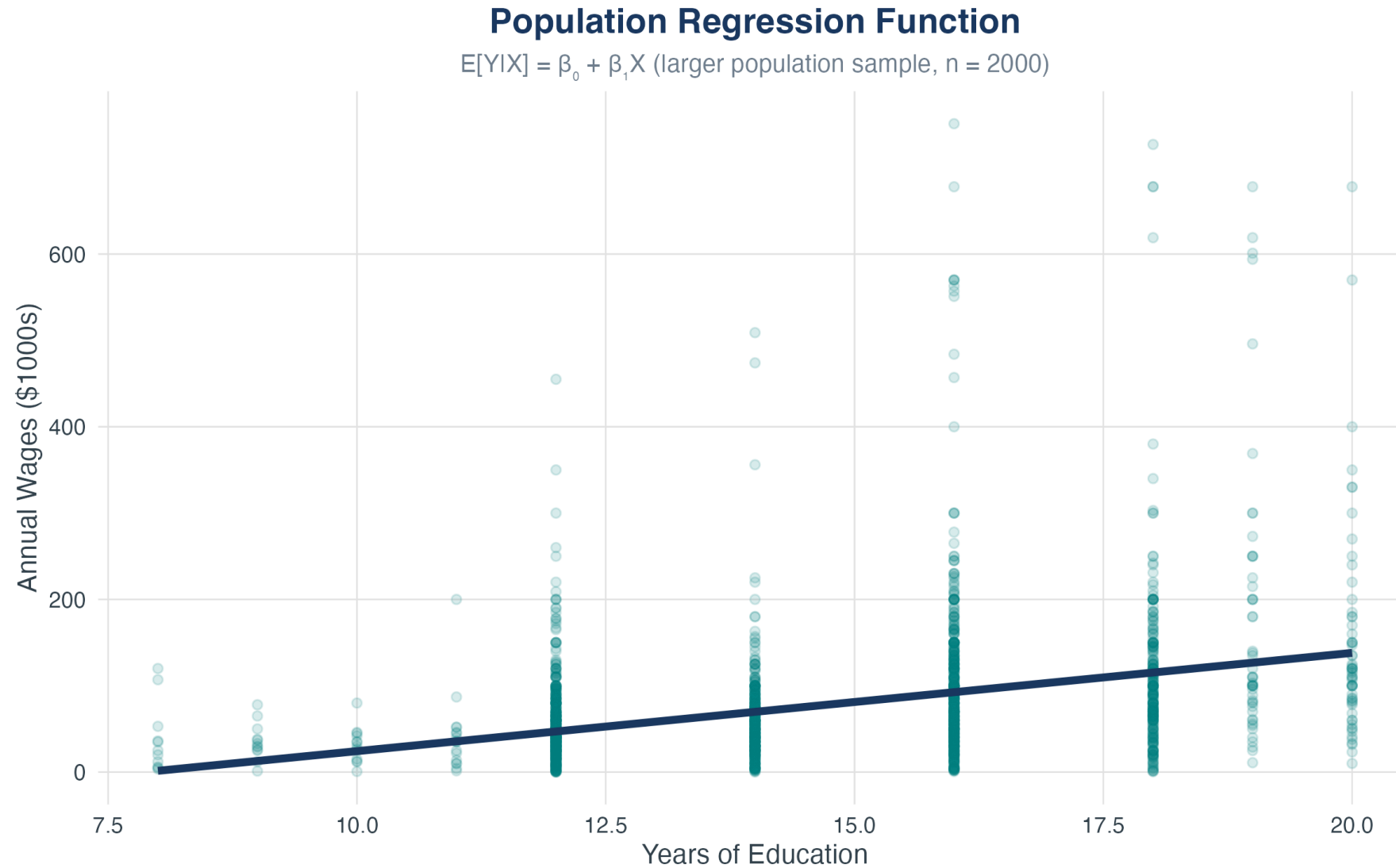
The mean of  $u$  is the same at every value of  $X$ .

Combine this with the normalization  $E[u] = 0$ :

$$E[u|X] = 0$$

► This is our workhorse assumption. It implies the **population regression function** is linear:  $E[Y|X] = \beta_0 + \beta_1 X$

# Population Regression Function (PRF)



- The line shows  $E[Y|X] = \beta_0 + \beta_1 X$



# Assumption 2: i.i.d. Sampling

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## Independent and Identically Distributed (i.i.d.)

The observations  $(X_i, Y_i)$  are:

- **Independent:** One observation doesn't affect another
- **Identically distributed:** All drawn from the same population

## Why this matters:

- Ensures our sample is representative
- Rules out things like: time series (where today depends on yesterday), clustering (where observations within groups are related)
- Allows us to apply the Law of Large Numbers and Central Limit Theorem

## When violated:

- Standard errors are wrong (we'll cover this in later chapters)

# Assumption 3: No Extreme Outliers

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## Finite Fourth Moments

$$E[X^4] < \infty \text{ and } E[Y^4] < \infty$$

### In plain English:

- Extremely large values of  $X$  and  $Y$  are unlikely
- Ensures our estimators have well-defined variances
- Almost always satisfied in practice with real data

### Practical implication:

# Sampling Distributions and Uncertainty

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# $\hat{\beta}_1$ is a Random Variable

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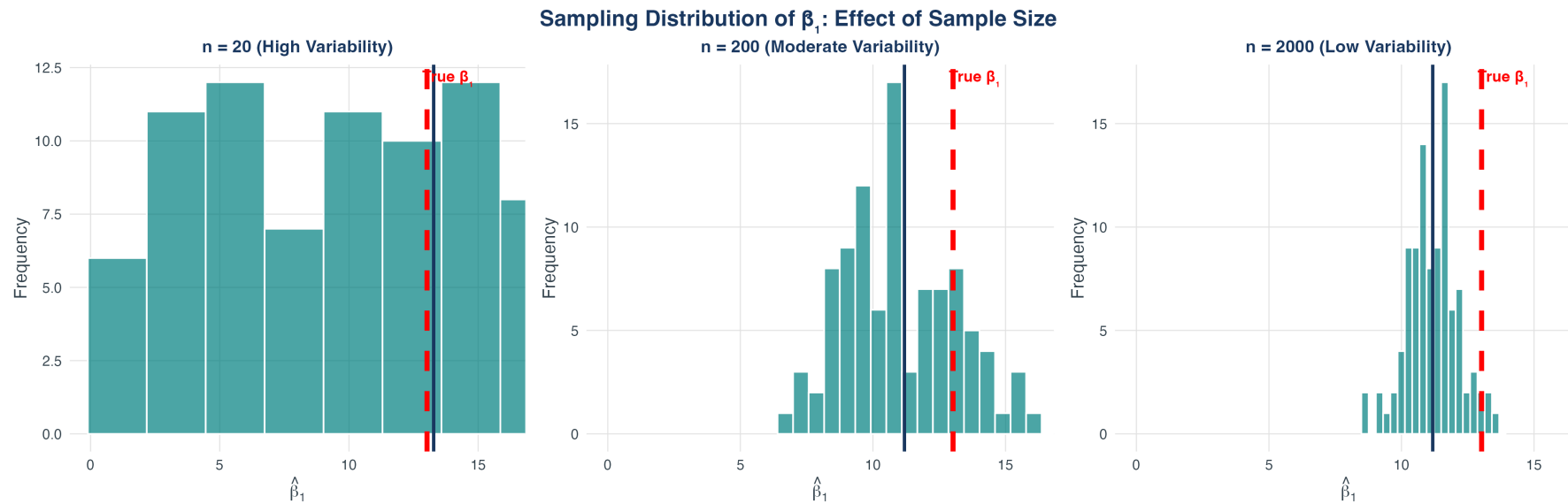
**Key insight:** Our estimates  $\hat{\beta}_0$  and  $\hat{\beta}_1$  depend on the sample.

- Different samples  $\Rightarrow$  different estimates
- If we drew a new sample, we'd get a different  $\hat{\beta}_1$
- This creates a **sampling distribution**

**Under the three assumptions:**

- $\hat{\beta}_1$  is **unbiased**:  $E[\hat{\beta}_1] = \beta_1$
- $\hat{\beta}_1$  is **consistent**: As  $n \rightarrow \infty$ ,  $\hat{\beta}_1 \rightarrow \beta_1$

# Sampling Distribution Visualization



- 1,000 different samples, each with  $n = 300$
- Each sample gives a different  $\hat{\beta}_1$
- Mean of estimates equals true  $\beta_1$  (unbiased!)
- Larger  $n \Rightarrow$  tighter distribution (more precise)

# Effect of Sample Size on Precision

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# What Determines the Variance of $\hat{\beta}_1$ ?

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The variance of  $\hat{\beta}_1$  is:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma_u^2}{n \cdot \text{Var}(X)}$$

Interpretation:

- **Larger  $n$** : More data  $\Rightarrow$  smaller variance (more precise)
- **Larger  $\text{Var}(X)$** : More spread in  $X$   $\Rightarrow$  smaller variance (easier to detect slope)
- **Larger  $\sigma_u^2$** : Noisier errors  $\Rightarrow$  larger variance (harder to estimate)

► We'll use this in the next chapter for hypothesis testing and confidence intervals.

# Common Misconceptions and Clarifications

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# Misconception #1: “Correlation = Causation”

## Wrong Thinking

“Education and wages are correlated, so education *causes* higher wages.”

## Correct Thinking

Regression shows **association**, not necessarily causation.

- Could be reverse causality (higher earners invest more in education)

...

- Could be omitted variable bias (ability affects both)

...

- Need experimental or quasi-experimental design for causal claims

# Misconception #2: “High $R^2$ = Good Model”

## Wrong Thinking

“ $R^2 = 0.30$  is low, so this is a bad model.”

## Correct Thinking

$R^2$  depends on the context!

- In physics:  $R^2 > 0.95$  is common (physical laws are precise)
- ...
- In social sciences:  $R^2 = 0.30$  can be excellent (human behavior is complex)
- ...
- $R^2$  measures *explained variance*, not *model validity*

# Misconception #3: “Intercept Must Be Meaningful”

## Wrong Thinking

“ $\beta_0 = -114.7$  means someone with zero education earns -\$114,700? That’s impossible!”

## Correct Thinking

The intercept is often **not interpretable** in practice.

- It’s the predicted  $Y$  when  $X = 0$
- ...
- But  $X = 0$  may be outside the range of data (no one has 0 years of education)
- ...
- It’s an *extrapolation*, not an actual prediction

# Misconception #4: “Residuals = Errors”

## Wrong Thinking

“The residual  $\hat{u}_i$  is the same as the true error  $u_i$ .”

## Correct Thinking

They are **different** (and we never observe  $u_i$ !):

- **Error**  $u_i = Y_i - (\beta_0 + \beta_1 X_i)$  (population, unknown)

...

- **Residual**  $\hat{u}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$  (sample, observed)

# Misconception #5: “Adding X Always Helps”

## Wrong Thinking

“More variables = better model, always.”

## Correct Thinking

Only add variables that **belong** in the model:

- Irrelevant variables increase standard errors (reduce precision)

...

- Can lead to multicollinearity problems

...

- Overfitting: model fits sample noise, not population relationship

# Implementing OLS in Stata

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# Complete Stata Workflow

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## Full analysis in Stata:

```
* Load data
import delimited "education_wages_data.csv", clear

* Summary statistics
summarize education wages

* Scatter plot
scatter wages education

* Correlation
correlate education wages

* Run OLS regression
regress wages education

* Generate fitted values and residuals
predict wages_fitted, xb
predict residual, residuals
```

# Calculating Fit Statistics in Stata (Part 1)

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Calculate TSS, ESS, and SSR manually:

```
* After running: regress wages education
* Store basic stats from regression
scalar r2 = e(r2)
scalar ser = e(rmse)
scalar n = e(N)

* Calculate mean of Y
quietly sum wages
scalar y_bar = r(mean)

* Generate squared deviations
gen deviation_total = (wages - y_bar)^2
gen deviation_explained = (wages_fitted - y_bar)^2
gen deviation_residual = residual^2
```

**What we're doing:** Creating three variables to capture total, explained, and unexplained variation.



# Calculating Fit Statistics in Stata (Part 2)

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Sum up and verify the decomposition:

```
* Sum the squared deviations
quietly sum deviation_total
scalar TSS = r(sum)

quietly sum deviation_explained
scalar ESS = r(sum)

quietly sum deviation_residual
scalar SSR = r(sum)

* Verify: TSS = ESS + SSR
display "TSS = " TSS
display "ESS = " ESS
display "SSR = " SSR
display "Check: TSS - (ESS + SSR) = " (TSS - (ESS + SSR))
```

**Interpretation:** The check should equal zero (or very close), confirming  $TSS = ESS + SSR$ .

# Diagnostic Plots in Stata

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```
* Scatter with regression line
scatter wages education || lfit wages education, ///
legend(off) title("OLS Regression Line")

* Residual plot
scatter residual wages_fitted, ///
yline(0) ///
title("Residual Plot") ///
ytitle("Residuals") xtitle("Fitted Values")

* Histogram of residuals
histogram residual, normal ///
title("Distribution of Residuals")
```

**Full walkthrough script:** `econ3500_ols_walkthrough.do`

# Bringing It All Together

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# The Big Picture

## ! From Data to Inference

Today's journey:

1. **Observed** relationships in scatter plots
2. **Quantified** them with OLS regression
3. **Assessed** fit with  $R^2$ , SER, and decomposition
4. **Understood** what assumptions let us interpret  $\beta_1$  causally
5. **Recognized**  $\hat{\beta}_1$  varies across samples (sampling distribution)

**Next chapter:** Hypothesis testing and confidence intervals — how to make formal inferences about  $\beta_1$ .

# Key Takeaways

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1. OLS gives the best linear fit by minimizing squared residuals
2. Slopes have precise interpretations:  $\Delta Y$  per unit  $\Delta X$ , holding  $u$  fixed
3.  $R^2$  and SER measure fit, but don't prove causality
4. Zero conditional mean is THE critical assumption for causal interpretation
5.  $\hat{\beta}_1$  has a sampling distribution — it's random, not fixed
6. Larger samples and more variation in  $X$  give more precise estimates

# Practice & Resources

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## Recommended practice:

- Stock & Watson Chapter 4 exercises
- **Work through:** `econ3500_ols_walkthrough.do`
- Try running regressions on the ACS data
- Practice interpreting coefficients out loud

## The Stata walkthrough includes:

- Complete worked example with education-wages data
- Manual calculation of all fit statistics
- Diagnostic plots
- Real-world application with ACS wage data
- Practice problems with solutions